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Task # 01

Raising a number n to a power p is the same as multiplying n by itself p times. Write a function called power() that takes a value for n and an int value for p , and returns the result as a double value.

CODE:

```java
import java.util.Scanner;
public class Main {

    // Function to calculate power
    public static double power(double n, int p) {
        double result = 1;
        for (int i = 0; i < p; i++) {
            result *= n;
        }
        return result;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number to be powered: ");
        double base = sc.nextDouble();
        sc.nextLine();

        System.out.print("Enter the number of times to be powered: ");
        int exponent = sc.nextInt();
        sc.nextLine();

        // Calling the power function
        double result = power(base, exponent);
        System.out.println(base + " raised to the power of " + exponent + " is " + result);
    }
}
```

OUTPUT:

```
Enter the number to be powered: 2
Enter the number of times to be powered: 3
2.0 raised to the power of 3 is 8.0
```

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Task # 02

CODE:

```
public class Main {

  // Overloaded function to calculate power for int
  public static double power(int n, int p) {
    double result = 1;
    for (int i = 0; i < Math.abs(p); i++) {
      result *= n;
    }
    if (p < 0) {
      return 1 / result;
    }
    return result;
  }

  // Overloaded function to calculate power for long
  public static double power(long n, int p) {
    double result = 1;
    for (int i = 0; i < Math.abs(p); i++) {
      result *= n;
    }
    if (p < 0) {
      return 1 / result;
    }
    return result;
  }

  // Overloaded function to calculate power for float
  public static double power(float n, int p) {
    double result = 1;
    for (int i = 0; i < Math.abs(p); i++) {
      result *= n;
    }
    if (p < 0) {
      return 1 / result;
    }
    return result;
  }

  public static void main(String[] args) {
    // Test int overload
    int intBase = 2;
    int intExponent = 3;
    double intResult = power(intBase, intExponent);
    System.out.println(intBase + " raised to the power of " +
intExponent + " is " + intResult);

    // Test long overload
    long longBase = 5L;
    int longExponent = 4;
    double longResult = power(longBase, longExponent);
    System.out.println(longBase + " raised to the power of "
+ longExponent + " is " + longResult);

    // Test float overload
    float floatBase = 2.5f;
    int floatExponent = 2;
    double floatResult = power(floatBase, floatExponent);
    System.out.println(floatBase + " raised to the power of "
+ floatExponent + " is " + floatResult);
  }
}
```

OUTPUT:

```
2 raised to the power of 3 is 8.0
5 raised to the power of 4 is 625.0
2.5 raised to the power of 2 is 6.25
```

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Task # 03

Simulate a simple library system. Create classes for Library, Book, and LibraryMember. TheLibrary class should contain a list of books. Write methods to check out and return books, and display library member information.

CODE:

```
import java.util.ArrayList;
import java.util.List;

// Class to represent a Book
class Book {
   String title;
   boolean isAvailable;

   // Constructor
   public Book(String title) {
      this.title = title;
      this.isAvailable = true;
   }

   // Method to check out a book
   public void checkOut() {
      if (isAvailable) {
         isAvailable = false;
         System.out.println("You have checked out: " + title);
      } else {
         System.out.println(title + " is already checked out.");
      }
   }

   // Method to return a book
   public void returnBook() {
      isAvailable = true;
      System.out.println("You have returned: " + title);
   }
}

// Class to represent a LibraryMember
class LibraryMember {
   String name;

   // Constructor
   public LibraryMember(String name) {
      this.name = name;
   }

   // Method to display member information
```

```
// Class to represent the Library
class Library {
   List<Book> books;

   // Constructor
   public Library() {
      books = new ArrayList<>();
   }

   // Method to add a book to the library
   public void addBook(Book book) {
      books.add(book);
   }

   // Method to display all books in the library
   public void displayBooks() {
      System.out.println("Books in Library:");
      for (Book book : books) {
         System.out.println(book.title + (book.isAvailable ? "
(Available)" : " (Checked out)"));
      }
   }
}

// Main class to simulate the library system
public class Main {

   public static void main(String[] args) {
      // Creating a library
      Library library = new Library();
      library.addBook(new Book("1984"));
      library.addBook(new Book("To Kill a Mockingbird"));

      // Creating library members
      LibraryMember member1 = new LibraryMember("Alice");
      LibraryMember member2 = new LibraryMember("Bob");

      library.displayBooks();

      member1.displayInfo();
      library.books.get(0).checkOut();

      member2.displayInfo();
      library.books.get(1).checkOut();
```

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```
    public void displayInfo() {
      System.out.println("Library Member: " + name);
    }
}
```

```
        member1.displayInfo();
        library.books.get(0).returnBook();

        member2.displayInfo();
        library.books.get(1).returnBook();

    }
}
```

OUTPUT:

```
Books in Library:
1984 (Available)
To Kill a Mockingbird (Available)
Library Member: Alice
You have checked out: 1984
Library Member: Bob
You have checked out: To Kill a Mockingbird
Library Member: Alice
You have returned: 1984
Library Member: Bob
You have returned: To Kill a Mockingbird
```

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Task # 04

Create overloaded methods for calculating the area of different shapes (circle, rectangle, triangle) and call them from the main method.

CODE:

```
public class Main {

   // Method to calculate the area of a circle
   public static double calculateArea(double radius) {
      return Math.PI * radius * radius; // Area of a circle: π * r²
   }

   // Method to calculate the area of a rectangle
   public static double calculateArea(double length, double width) {
      return length * width; // Area of a rectangle: length * width
   }

   // Method to calculate the area of a triangle using base and height
   public static double createArea(double base, int height) {
      return 0.5 * base * height; // Area of a triangle: 1/2 * base * height
   }

   public static void main(String[] args) {
      // Call overloaded methods from the main method
      double circleRadius = 5.0;
      double rectangleLength = 4.0;
      double rectangleWidth = 6.0;
      double triangleBase = 3.0;
      int triangleHeight = 7;

      // Calculate areas
      double circleArea = calculateArea(circleRadius);
      double rectangleArea = calculateArea(rectangleLength, rectangleWidth);
      double triangleArea = createArea(triangleBase, triangleHeight);

      // Display the results
      System.out.println("Area of the circle: " + circleArea);
      System.out.println("Area of the rectangle: " + rectangleArea);
      System.out.println("Area of the triangle: " + triangleArea);
   }
}
```

OUTPUT:

```
Area of the circle: 78.53981633974483
Area of the rectangle: 24.0
Area of the triangle: 10.5
```